Faculty of Engg. & Technology, Agra College, Agra

Foundation of Computer Science

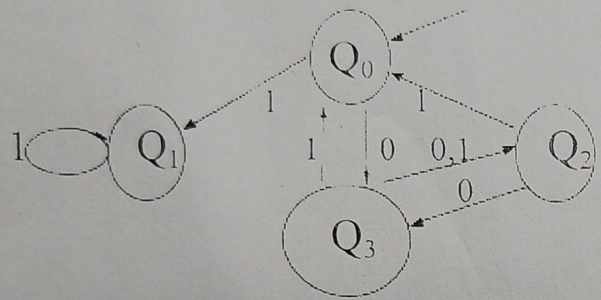
CLASS TEST-2

ROLL NUMBER..

COURSE	:	M.TECH	SUBJECT	:	FOC
YEAR	:	FIRST YEAR	CODE	:	MTCS101
SEMESTER	:	II	TIME	:	1.00 HRS
BRANCH	:	CS	MAX MARKS	:	20

Note : Attempt Any Four questions:

Q1:- What is DFA? Check whether the following machine is in DFA or NFA? If NFA then find DFA?



Q2:- Minimize the following automata?

State	Input	
	0	1
→qo	q1	q5
q1	q6	q2
q2	qo	q2
q3	q2	q6
q4	q7	q5
q5	q2	q6
q6	q6	q4
q7	q6	q2

Q3:- What is grammar? Find the L(G) from the G=({S},{a},{S→ aS, S → Λ},S)

Q4:- Find G from L(G)={ $a^n b^n c^i$, where $n \geq 1$, $i \geq 0$} ?

Q5:- Give any example for DFA accepting any string w? Also give accepting sequence?

Faculty of Engineering & Technology
Agra College, Agra
Class Test -II

Subject Name: Advance Algorithm
Branch :M.Tech CS
Year/Semester: 1st/1st

Subject Code: MTCS-102
Duration: 1 hour
Maximum Marks: 20

Note: Attempt any four questions. All questions carry equal marks (5X4)

Q1. Explain Merge sort with suitable example

Q2. Create B-Tree with order 5 from following data items 20, 30, 35, 85, 10, 55, 60, 25, 5, 65, 70, 75, 15, 40, 50, 80, 45.

Q3. Explain theory of NP Completeness.

Q4. Explain BFS with algorithm and suitable example.**Q5.** What is dynamic programming? Explain with suitable example.

Q6. Explain greedy algorithm with suitable example.

Faculty of Engineering & Technology
Agra College, Agra
Class Test -2

Subject Name: Research Process and Methodology
Branch :M.Tech CS
Year/Semester: 1st/1st

Subject Code: MTCC-101
Duration: 1 hour
Maximum Marks: 20

Note: Attempt any four questions. All questions carry equal marks **(5X4)**

Q1. Define classification of data in research and outline its primary objectives. Differentiate between geographical, chronological, qualitative, and quantitative bases of classification with one example each.

Q2. Describe observation as a data collection method, including structured vs. unstructured approaches. What are key challenges in ensuring reliability through this method?

Q3. Define sampling in research and explain the difference between probability and non-probability sampling. When would stratified random sampling be more appropriate than simple random sampling?

Q4. What is data analysis in research? Distinguish between descriptive, inferential, and predictive analysis, providing one statistical technique example for each.

Q5. Define a research hypothesis, differentiating null (H0) from alternative (H1). Outline the steps in hypothesis testing, including Type I and Type II errors.

Q6. Explain statistical inference, including point estimation and interval estimation. How does a 95% confidence interval relate to hypothesis testing outcomes?

Q7. What factors influence the interpretation of statistical results (e.g., p-value, effect size, practical significance)? Why is effect size more informative than p-value alone in real-world research?

F.E.T. AGRA COLLEGE, AGRA

M M : 20

CLASS TEST # 2

TIME : 1 HOUR

NEURAL NETWORKS

MTCS-026 M.Tech. [CS - I SEM] 2025

NOTE : ATTEMPT ANY 4 QUESTIONS. ALL QUESTIONS CARRY EQUAL MARKS.

Q(1): Discuss the role of Artificial Neural Networks in Machine Learning ? Discuss various Areas of Application of Artificial Neural Networks ?

Q(2): Explain Artificial Neural Networks ? Discuss the Architecture of following ANN's ?

- Hebb's Neural Network
- Perceptron Neural Network
- Backpropagation Neural Networks

Q(3): Differentiate between Feed Forward ANN and Recurrent ANN with the help of suitable diagram ?

Q(4): Differentiate between Classification and Clustering with the help of suitable example ?

Q(5): Differentiate among Supervised Learning, Unsupervised Learning and Reinforced Learning along with suitable diagrams and examples ?

Q(6): Differentiate between Pattern Classification and Pattern Association via suitable examples ?

Q(7): Design an ANN for Binary NAND function via threshold approach?

Q(8): Design an ANN for Binary NOR function via bias approach ?

Q(9): Design an ANN for Binary EX - NOR function via threshold ?

Q(10): Discuss various Learning Rules / Learning used in ANN ?

Faculty of Engineering and Technology

Software Project Management (CT2)

M Tech(CS) I year

MM 20 TIME 1 HOURS

Attempt any three Questions. Question 1 is compulsory

Que1 what is testing? Explain the various types of Testing.(8)

Que2What is process? And explain the various process elements.(6)

Que3 what is estimation? Explain the Estimation approach.(6)

Que 4 What is WBS? Explain in details.(6)